Durham University Business School

Investment decisions - Project Appraisal

Dr Lilly Xinyi Huang

AACSB ACCREDITED · ASSOCIATION OF MBAS ACCREDITED · EFMD EQUIS ACCREDITED

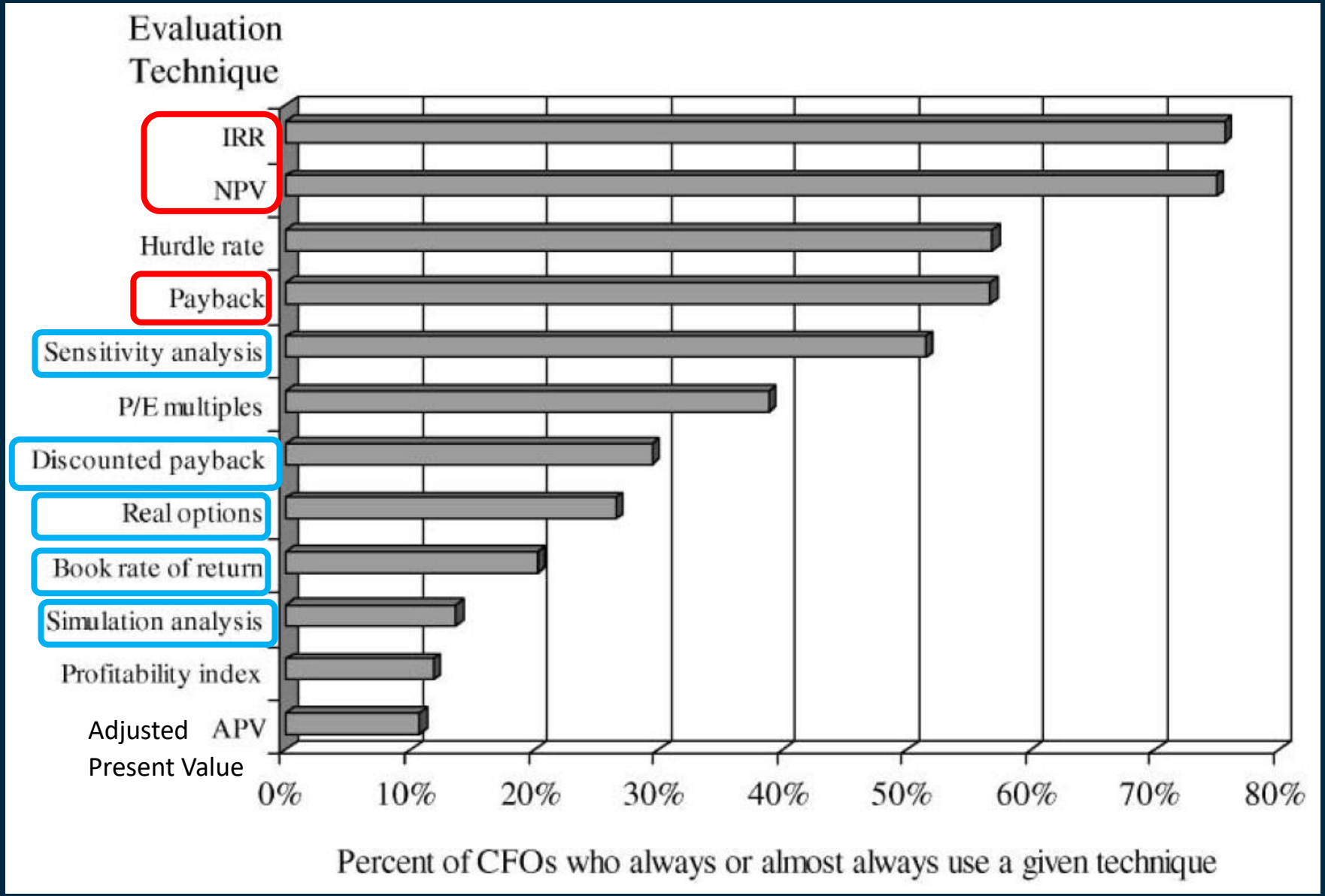
Poll Everywhere Audience Response System

- Using your mobile, tablet or laptop:
  - Get the Poll Everywhere App from Google Play or AppStore.
  - If you don't want to use apps go to: PollEv.com/xinyihuang180

At the end of this session you should have

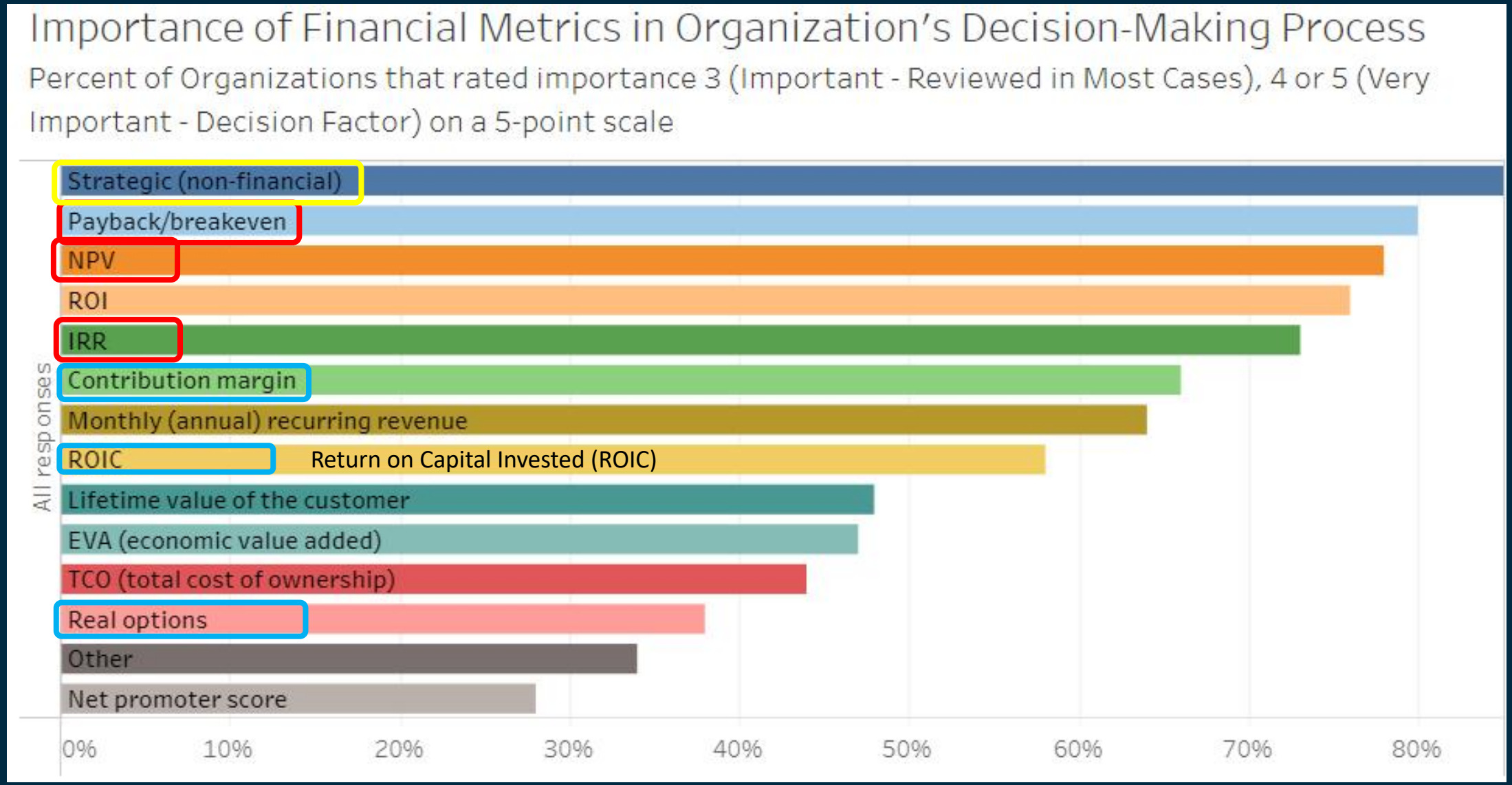
- built upon your knowledge from the Foundations of finance with regards to Discounting Cash Flows and Investment Appraisal Methods to extend you knowledge and skills.
- The ability to use of a wider range of investment appraisal methods, including
  - ✓Accounting rate of return (ARR) and Return on Invested Capital.
- learned to appreciate the importance of Sensitivity Analysis, Scenario Analysis and Break-even Analysis.
- be able to conduct analysis and appraisal for ongoing investment projects via decision trees and real option valuations.

Investment Appraisal Methods in the USA 2001



Source: Graham, J. and Harvey, C. (2001) The theory and practice of Corporate Finance: evidence from the field, *Journal of Financial Economics*, p. 197

Investment Appraisal Methods in the UK 2020



Source: Association for Financial Professionals (2020) 2020 AFP FP&A Survey: Project Investment and Cost of Capital Survey Dashboard.

Investment appraisal methods using accounting approaches

- Accounting Rate of Return (ARR)

$$\text{ARR} = \frac{\text{average annual net income (profit) over the investment period}}{\text{average investment}} * 100$$

$$\text{Average investment} = \frac{\textit{Initial value of investment} + \text{end value of the investment}}{2}$$

Birdbox Ltd wants to expand its production. This requires initial investment of £10,000 and is expected to generate accounting profits of £2,000 in year 1, £2,000 in year 2, £3,000 in year 3, £5,000 in year 4. The scrap is worth £1,000. What is the ARR?

0

The ARR would be 27.3% **(A)**

The ARR would be 218% **(B)**

The ARR would be 54.6% **(C)**

The ARR would be 109% **(D)**

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Birdbox Ltd wants to expand its production. This requires initial investment of £10,000 and is expected to generate accounting profits of £2,000 in year 1, £2,000 in year 2, £3,000 in year 3, £5,000 in year 4. The scrap is worth £1,000. What is the ARR?

0

(A) The ARR would be 27.3%

0%

(B) The ARR would be 218%

0%

(C) The ARR would be 54.6%

0%

(D) The ARR would be 109%

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Birdbox Ltd wants to expand its production. This requires initial investment of £10,000 and is expected to generate accounting profits of £2,000 in year 1, £2,000 in year 2, £3,000 in year 3, £5,000 in year 4. The scrap is worth £1,000. What is the ARR?

0

(A) The ARR would be 27.3%

0%

(B) The ARR would be 218%

0%

(C) The ARR would be 54.6%

0%

(D) The ARR would be 109%

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Investment appraisal methods using accounting approaches

- **Accounting Rate of Return (ARR)**

$$\text{ARR} = \frac{\text{average net income (profit) over the investment period}}{\text{average investment}} * 100$$

- Birdbox Ltd wants to expand its production. This requires an initial investment of £10,000 and is expected to generate accounting profits of £2,000 in year 1, £2,000 in year 2, £3,000 in year 3, £5,000 in year 4. The scarp value at the end is £1,000. What is the ARR?
- $\text{Average annual profits} = \frac{£2{,}000 + £2{,}000 + £3{,}000 + £5{,}000}{4} = £3{,}000$
- $\text{Average investment} = \frac{£10{,}000 + £1{,}000}{2} = £5{,}500$
- $\text{ARR} = \frac{£3{,}000}{£5{,}500} * 100\% = 54.6\%$
- Note: The initial costs of the investment of £10,000 is taken into account in the annual costs of the project via depreciation!

Investment appraisal methods using accounting approaches

- Problems with the Accounting Rate of Return (ARR)
  - ✓ Average investment: The ARR <u>increases</u> if the firm gives away the scrap rather than charging £1,000.
  - ✓ $\text{Average investment} = \frac{£10{,}000}{2} = £5{,}000$
  - ✓ $\text{ARR} = \frac{£3{,}000}{£5{,}000} * 100\% = 60.0\%$
  - ✓ A project with a higher ARR can make less profit on the same initial capital expenditure and scrap value.

- **Return on Capital Invested (ROIC)**

$$Return\ on\ Invested\ Capital\ (ROIC) = \frac{Net\ Operating\ Profit\ after\ tax\ (NOPAT)}{(Book\ value\ of)\ Invested\ Capital}$$

$$Return\ on\ Invested\ Capital\ (ROIC) = \frac{Net\ Operating\ Profit\ after\ tax\ (NOPAT)}{Sales} * \frac{Sales}{Invested\ Capital}$$

Recap from Foundations of Finance

- Investment appraisal methods using cash flow approaches
  - ✓Investment appraisal methods of NPV, IRR, payback period and real options focus on (net) cash flows, NOT on accounting costs and revenues!
  - ✓Need to consider how to treat inflation:
    - ➢If a nominal hurdle rate is used (i.e. the rate is increased for inflation), then nominal cash flows should also be used (i.e. adjusted for expected inflation).
    - ➢If a real hurdle rate is used, then real cash flows should be used.
- Payback period
  - ✓Project choice is based on how quickly the initial financial outlay is paid back.
  - ✓No consideration of income generated after pay-back date or net present value (NPV).
  - ✓Could we combine NPV and calculate discounted cash flow payback period?

Teetree Florists plans to purchase a stand to sell flowers at local markets. The stand costs £10,500, the net cash flow from sales CF in year 1 is expected to be £5,000, and £6,000 in years 2 and 3. R=10%. The PB period is 3 years max.

- Teetree Florists should not buy the stand as the payback period exceeds the maximum three years.
- Teetree Florists should buy the stand as the payback period is three years.
- Teetree Florists should buy the stand as the payback period is two years.
- It is impossible to calculate the discounted cash flow periods, as some of the relevant data is missing.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Teetree Florists plans to purchase a stand to sell flowers at local markets. The stand costs £10,500, the net cash flow from sales CF in year 1 is expected to be £5,000, and £6,000 in years 2 and 3. R=10%. The PB period is 3 years max.

Teetree Florists should not buy the stand as the payback period exceeds the maximum three years.	0%
Teetree Florists should buy the stand as the payback period is three years.	0%
Teetree Florists should buy the stand as the payback period is two years.	0%
It is impossible to calculate the discounted cash flow periods, as some of the relevant data is missing.	0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Teetree Florists plans to purchase a stand to sell flowers at local markets. The stand costs £10,500, the net cash flow from sales CF in year 1 is expected to be £5,000, and £6,000 in years 2 and 3. R=10%. The PB period is 3 years max.

Teetree Florists should not buy the stand as the payback period exceeds the maximum three years.	0%
Teetree Florists should buy the stand as the payback period is three years.	0%
Teetree Florists should buy the stand as the payback period is two years.	0%
It is impossible to calculate the discounted cash flow periods, as some of the relevant data is missing.	0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Discounted Payback Period

- Could we combine NPV and calculate discounted cash flow payback period?
  - ✓Teetree Florists plans to purchase a stand to sell flowers at local markets. The stand costs £10,500, the net cash flow from sales C in year 1 is expected to be £5,000, and £6,000 in years 2 and 3. R = 10%. The PB period is 3 years max.
  - ✓NPV Year 1: -£10,500 + (£5,000*$\frac{1}{(1+0.1)}$) = -£10,500 +£4,545.45 = -£5,954.55
  - ✓NPV Year 2: -£5,954.55 + (£6,000*$\frac{1}{(1+0.1)^2}$) = -£5,954.55 + 4,958.68 = -£995.87
  - ✓NPV Year 3: -£995.87 + (£6,000*$\frac{1}{(1+0.1)^3}$) = -£995.87 + 4,507. 89 = +£3,512.02

Net Present Value Method (NPV)

- Investment appraisal method using cash flow approach
  - ✓Investment project is evaluated based on the present value of cash in- and out-flows along the life-span of the investment.
  - ✓Requires the setting of a hurdle rates (e.g. cost of finance; expected returns) to calculate the discounts.
    - ➢Changes to the required rate of return can have a significant influence on which projects are viable, and which aren't.
    - ➢The higher the required rate of return, the more long-term investment options are penalised.
  - ✓Assumes that managers are reasonably confident of cash-in and cash-outflows.

Net Present Value Method (NPV): The Equivalent Annual Cost Method

- How to compare investments in e.g. machinery with different time-lines, where the machinery needs to be continuously available? E.g.
  - ✓Soft drinks manufacturers need to have continuously access to bottling plants as they otherwise are unable to sell their products.
  - ✓They might need to choose between machinery with different life-spans.
  - ✓They might need to take a decision at which point to replace the plant. (e.g. considering the cost of replacement versus and benefits of improved efficiency of the new machinery).
- Equivalent Annual Cost Method

The Equivalent Annual Cost Method: Charity gym

- A charity gym wants to lease a new weightlifting machine and has received offers from two different leasing firms. As the machines do not generate any revenues, at this point we will consider only cash outflows.
- Weightmax (W), costs an up-front leasing fee of £400, and a fixed maintenance and rent fee of £100 must be paid at the end of each year. The contract lasts for three years, and at the end the leasing company will come and collect the machine.
- Powerplay (P), costs an up-front leasing fee of £500, and a fixed maintenance and rent fee of £80 must be paid at the end of each year. The contract lasts for four years, and at the end the leasing company will come and collect the machine.
- Calculate the NPV (net present value) of the two options if the hurdle rate if the Board of Trustees sets is 8%.

Write down the NPV (remember the sign) for W and P, by writing e.g. + £345.78P and - £23.23W

0

Nobody has responded yet.

Hang tight! Responses are coming in.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

The Equivalent Annual Cost Method: Charity gym

- $\text{NPV} = C_0 + \frac{C_1}{1 + r_1}$
- Weightmax (W): $\text{NPV}_W = -£400 + \frac{(-£100)}{1.08} + \frac{(-£100)}{1.08^2} + \frac{(-£100)}{1.08^3}$
  - ✓NPV_W = -400 - £92.59 - £85.73 – 79.38 = - £657.00
- Powerplay (P): $\text{NPV}_P = -£500 + \frac{(-£80)}{1.08} + \frac{(-£80)}{1.08^2} + \frac{(-£80)}{1.08^3} + \frac{(-£80)}{1.08^4}$
  - ✓NPV_P = -500 - £74.07 – 68.59 – 63.51 – 58.80 = -£764.97
- The NPV for Weightmax is lower, therefore this appears to be the better choice for the Charity, as it is less costly.
  - ✓However, this ignores that the Weightmax allows the weightlifting machine to be used only for 3 years and Powerplay one for four years.

The Equivalent Annual Cost Method: Charity gym

- Due to differences in the lifetime of investments, firms are interested in the **equivalent annual cost of machines**.
- This calculation draws on the annuity calculations discussed in FoF.
- Regarding the total value of the cash outflows, firms should be indifferent whether they pay a lump sum of £657.00 (or £764.97) at the beginning of the period, or 3 (4) equal payments across the period, if the values are properly discounted.
- $NPV\ of\ an\ annuity = Annuity\ Payment * Annuity\ factor = C * \left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right)$
- So, we can calculate the annuity payment C as:

✓ $C = \frac{NPV}{\frac{1-\frac{1}{(1+r)^t}}{r}}$ $or\ C = \frac{NPV}{\left(\frac{1}{r} - \frac{1}{r(1+r)^t}\right)}$

✓ $A_W = \frac{£657}{\frac{1-\frac{1}{(1+0.08)^3}}{0.08}} = \frac{£657}{2.5771} = £254.94$ and $A_P = \frac{£764.97}{\frac{1-\frac{1}{(1+0.08)^4}}{0.08}} = \frac{£764.97}{3.3121} =$

£245.10

Net Present Social Value (NPSV)

- Especially in relation to large-scale infrastructure projects, sponsors try to value not only cash flows but also non-financial impacts of the investment, such as
  - ✓Reduction (increase) in pollution (e.g. CO2 emissions) v. habitat loss.
  - ✓Identification v. destruction of historical sites.
  - ✓Wider impact on local, regional, national or international economic growth.
  - ✓Wider impact on skills development and social fairness or mobility.
- Challenges:
  - ✓What is the appropriate hurdle rate?
    - ➢The higher the hurdle rate the more future costs or benefits are discounted.
    - ➢How does this affect future generations?
  - ✓How are non-commercial goods and services valued?

Internal Rate of Return (IRR) Method

- Investment appraisal method using cash flow approach.

Which of the following statements about the IRR method of investment appraisal are correct?

- One key benefit of the IRR method of investment appraisal is that it's validity doe…
- IRR can be calculated by setting up a normal NPV calculation but then setting the …
- Depending on the distribution of positive and negative cash flows over time, the I…
- An approximate value of IRR can often be calculated by interpolation between tw…
- IRR is a superior method for investment appraisal compared to NPV.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following statements about the IRR method of investment appraisal are correct?

One key benefit of the IRR method of investment appraisal is that it's validity does not depend on the accuracy of managers' forecasts of expected cash flows.

0%

IRR can be calculated by setting up a normal NPV calculation but then setting the NPV as zero.

0%

Depending on the distribution of positive and negative cash flows over time, the IRR calculation can have more than one solution.

0%

An approximate value of IRR can often be calculated by interpolation between two assumed return values.

0%

IRR is a superior method for investment appraisal compared to NPV.

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Which of the following statements about the IRR method of investment appraisal are correct?

One key benefit of the IRR method of investment appraisal is that it's validity does not depend on the accuracy of managers' forecasts of expected cash flows.

0%

IRR can be calculated by setting up a normal NPV calculation but then setting the NPV as zero.

0%

Depending on the distribution of positive and negative cash flows over time, the IRR calculation can have more than one solution.

0%

An approximate value of IRR can often be calculated by interpolation between two assumed return values.

0%

IRR is a superior method for investment appraisal compared to NPV.

0%

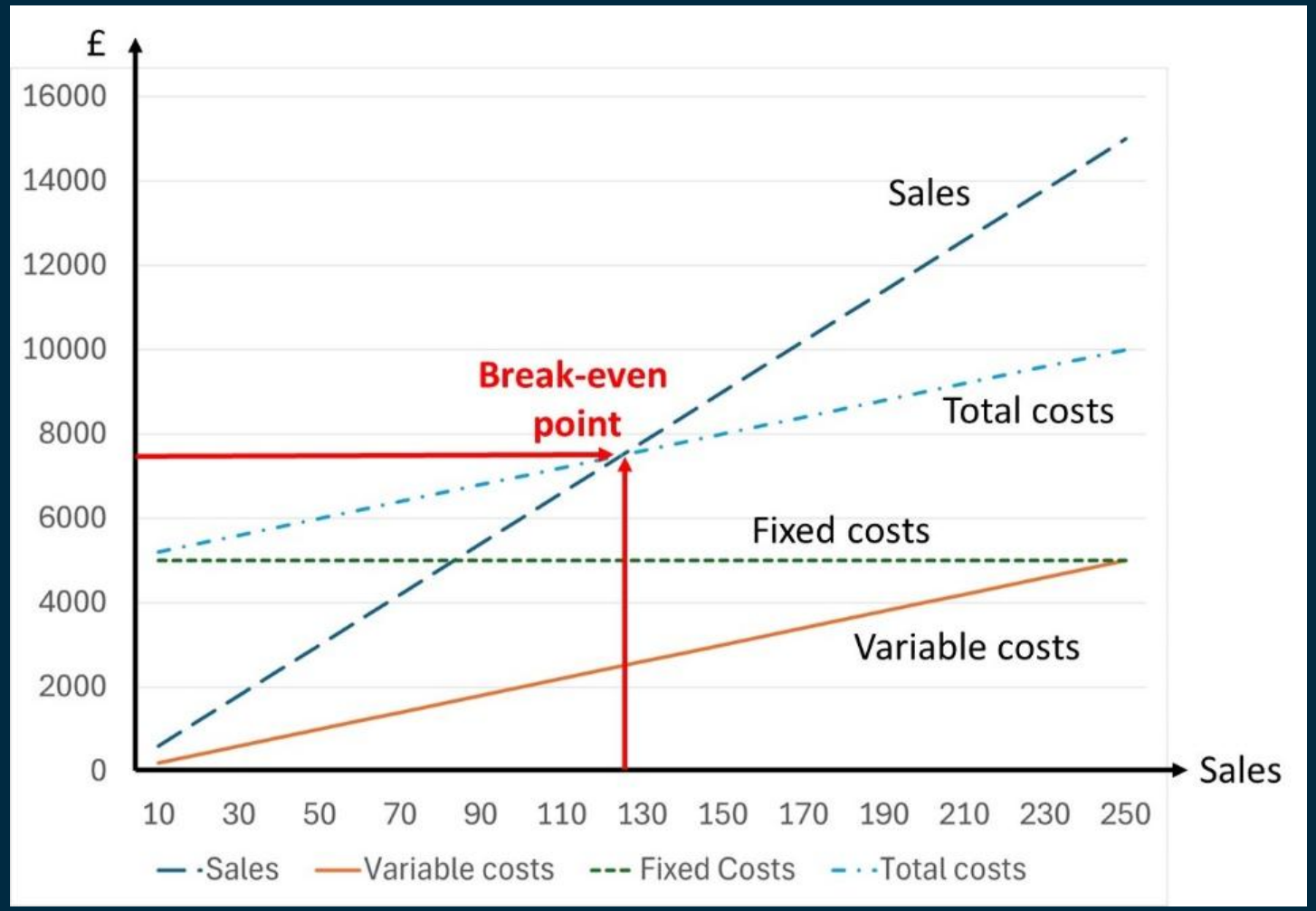
Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

From Sensitivity Analysis to Scenario Analysis

- Sensitivity Analysis highlights forecast uncertainty and allows managers to consider
  - ✓Whether/how the firm will cope if expectations are not met.
  - ✓Whether the firm can mitigate risks
    - ➢e.g. leasing instead of buying an asset with an option of purchase at a later date.
- Limitation: **Sensitivity Analysis** considers the implications of ONE factor developing worse than expected at a time.
- **Scenario Analysis** considers a scenario affecting MULTIPLE factors.
  - ✓E.g. Stress Tests for Financial Institutions.

Dealing with Uncertainty: Break-even Analysis

- Though often used in accounting, break-even analysis can also be used in cash flow analysis.
  - ✓Identifies the sales at which the firm makes no "profits" or losses.



- ✓In contrast to accounting, in finance and economics all costs of capital are considered (via the discount rate), rather than just the cost of debt capital.
- ✓While from an accounting perspective up-front investment is spread across time (depreciation), from a cash flow perspective, it is considered at the outset.

Break-even Analysis using Accounting Ratios

- Typically starts with the calculation of the **pre-tax contribution margin**
  - ✓Contribution of each sale to the project's fixed costs (incl. depreciation)
  - ✓"Pre-tax" as no taxes due if the project makes no profits (Sales exclude VAT-type taxes)

$$Pre - tax\ contribution\ margin = Sales\ price\ per\ unit - variable\ costs\ per\ unit$$

- The number of units needed to be sold to reach the break-even point can be calculated as:

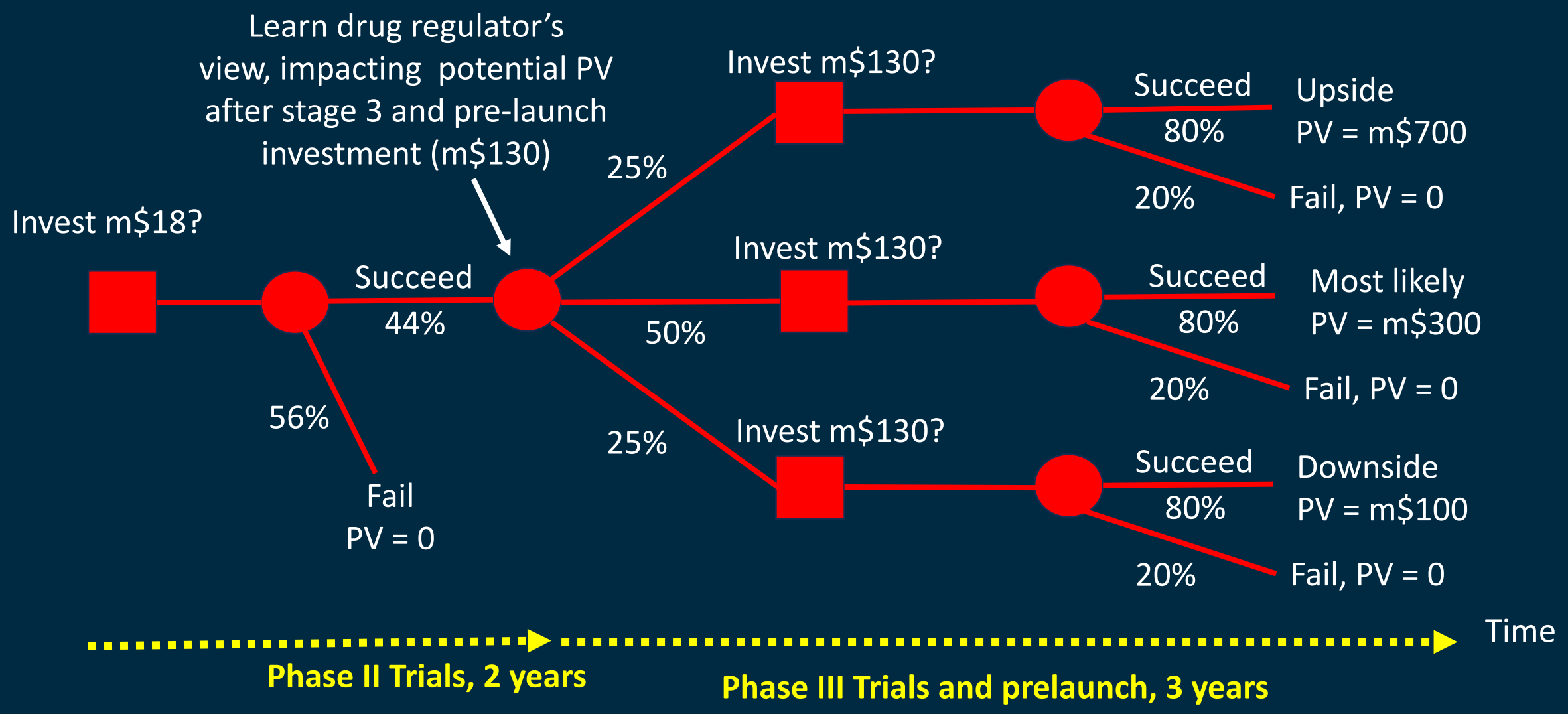
$$Break - even\ quantity = \frac{Fixed\ costs + Depreciation}{Pre - tax\ contribution\ margin} = \frac{Fixed\ costs + Depreciation}{Sales - Variable\ costs}$$

- Businesses experience variations in sales, e.g. due to seasonality or the business cycle. This leads to variations in average fixed costs per unit and variations in profits per unit: **Degree of operating leverage of a firm (DOL)**.

$$DOL = \frac{percentage\ change\ in\ profits}{percentage\ change\ in\ sales}$$

Ongoing management of investment projects: Decision trees

- Decision trees highlight the potential for sequential decision making.



Source: Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) *Principles of Corporate Finance*, 14th ed., McGraw Hill: New York, p. 289.

Ongoing management of investment projects: Decision trees

- Decision trees are solved starting with the right-hand side:

$$Expected\ payoff\ (NPV) = (probability\ of\ scenario\ A * Payoff\ of\ scenario\ A) + (prob$$

$$Payoff\ of\ scenario\ B)$$ + etc.

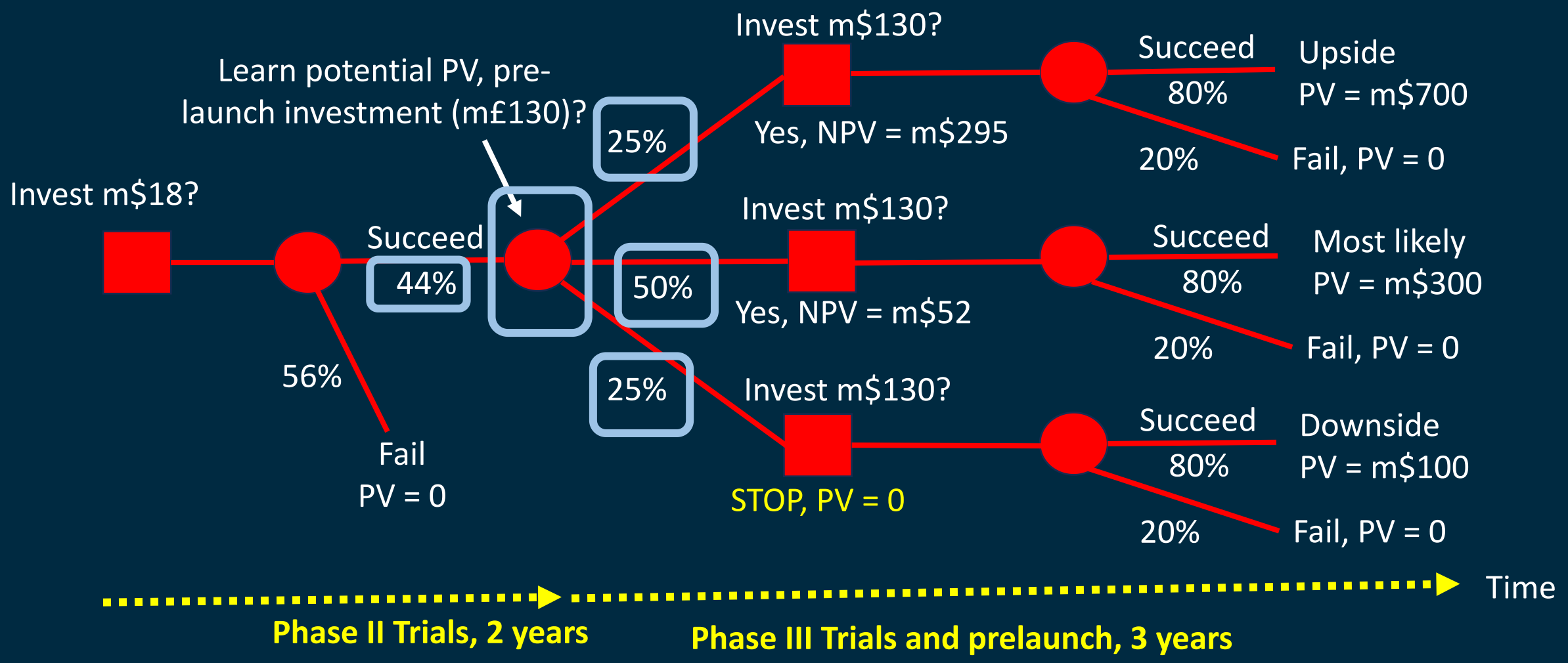
- The NPVs at the start of phase III trials are: (3 years, hurdle rate 9.6%):

$$NPV\ (upside) = -130 + 0.8 * \frac{700}{(1.096)^3} = \$295\ million$$ Yes

$$NPV\ (upside) = -130 + 0.8 * \frac{300}{(1.096)^3} = \$52\ million$$ Yes

$$NPV\ (upside) = -130 + 0.8 * \frac{100}{(1.096)^3} = -\$69\ million$$ No

E.g. Options to delay decisions: Decision trees



Source: Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) *Principles of Corporate Finance*, 14th ed., McGraw Hill: New York, p. 289.

Ongoing management of investment projects: Decision trees

- Decision trees are solved starting with the right-hand side:

$$Expected\ payoff\ (NPV\ or\ Profit) = (probability\ of\ scenario\ A * Payoff\ of\ scenario\ A) + (probability\ of\ scenario\ B * Payoff\ of\ scenario\ B)$$ + etc.

- **For stage 3 / prelaunch we can calculate the NPVs (3 years, hurdle rate 9.6%):**

$$NPV\ (upside) = -130 + 0.8 * \frac{700}{(1.096)^3} = \$295\ million$$ Yes

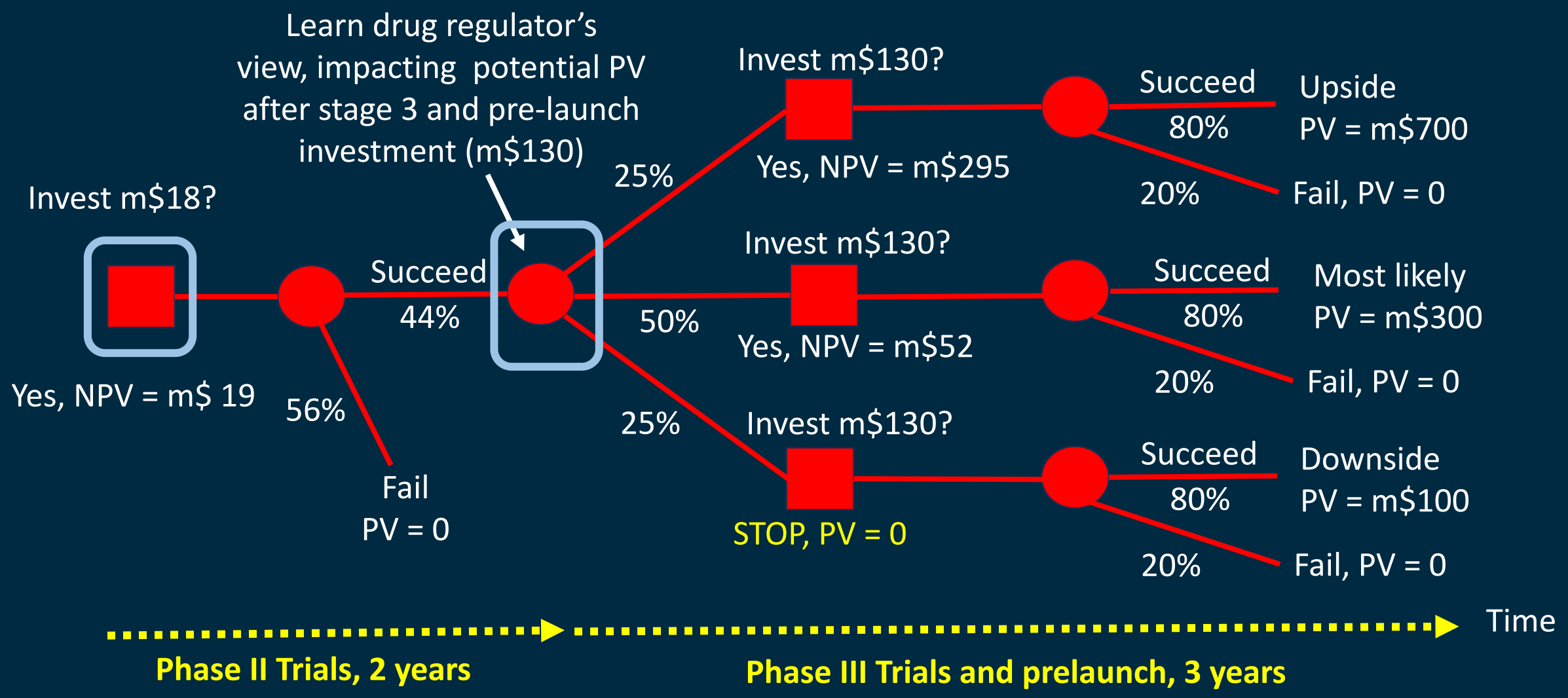
$$NPV\ (upside) = -130 + 0.8 * \frac{300}{(1.096)^3} = \$52\ million$$ Yes

$$NPV\ (upside) = -130 + 0.8 * \frac{100}{(1.096)^3} = -\$69\ million$$ No

- For stage 2 we can calculate the NPV (2 years, hurdle rate 9.6%):

$$NPV = -18 + 0.44 * \frac{0.25*295 + 0.5*52 + 0.25*0}{(1.096)^2} = -18 + 37 = \$19\ million$$ Yes

E.g. Option to delay decisions: Decision trees



Source: Brealey, R., Myers, S., Allen, F. and Edmans, A. (2022) *Principles of Corporate Finance*, 14th ed., McGraw Hill: New York, p. 289.

Ongoing management of investment projects: Decision trees

Decision trees

- can be used to consider the value of taking strategic decisions sequentially over time.
- still rely on managers to have a reasonably good idea about likely outcomes.
- to keep them manageable managers must identify key factors or scenarios which are likely to affect the project's outcomes.
- might highlight the benefit of creating options to take decisions at a later time – manage investment projects actively over time
  - ✓e.g. decisions about whether to buy an asset or to lease it with the option to purchase it later.
  - ✓Leads to Real Options Theory

Real Option Theory: Thinking strategically about investment decisions

- Traditional Discounted Cash Flow methods of investment appraisal (NPV, IRR) assume that managers have ex-ante knowledge about expected future cash flows.
  - ✓Passive investment strategy once the decision to invest has been taken.
- Real Options Theory suggests that there is a value to flexibility which allows decision markers to respond to different or changing conditions.
  - ✓As flexibility has a value to the firm, managers should consider designing projects such that they are able to respond flexibly.
  - ✓Valuing the benefits of this flexibility is estimated based on financial option valuation models (typically Black-Scholes or Binomal).
- E.g. proof-of-concept investments are not valued merely on the potential for success of this particular venture, but on the option the success of the investments offer to expand the project.

How much do you know about financial options? Which if the option below fits you best?

- I know in principle how financial options work, but I don't know any of the option…
- I know how financial options work, and I am able to value options using the bino…
- I know how financial options work, and I am able to value options using the Black…
- I know how financial options work, and I am able to value options using both the …
- I don't really now much about options.

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How much do you know about financial options? Which if the option below fits you best?

I know in principle how financial options work, but I don't know any of the option valuation methods.

0%

I know how financial options work, and I am able to value options using the binomial option valuation method.

0%

I know how financial options work, and I am able to value options using the Black-Scholes option valuation method.

0%

I know how financial options work, and I am able to value options using both the Black-Scholes and the binomial option valuation methods.

0%

I don't really now much about options.

0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

How much do you know about financial options? Which if the option below fits you best?

I know in principle how financial options work, but I don't know any of the option valuation methods.

0%

I know how financial options work, and I am able to value options using the binomial option valuation method.

0%

I know how financial options work, and I am able to value options using the Black-Scholes option valuation method.

0%

I know how financial options work, and I am able to value options using both the Black-Scholes and the binomial option valuation methods.

0%

I don't really now much about options.

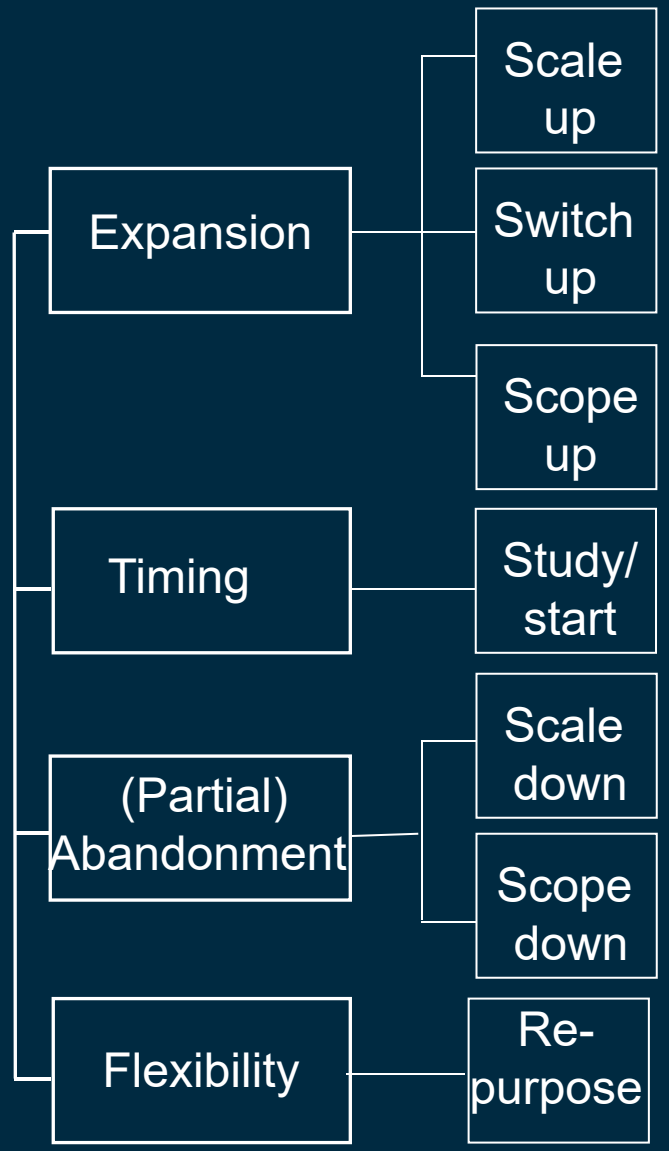
0%

Start the presentation to see live content. For screen share software, share the entire screen. Get help at **pollev.com/app**

Real Option Theory

- "Options are rights without obligations. An option is the right (but not an obligation) to buy or sell an asset at some point within a predetermined period of time for a predetermined price" (Copeland and Keenan 1998: 40).
  - ✓Call option: Right to buy an asset.
    - ➢If the exercise price is below the market price, the option is "in the money" and will be used.
  - ✓Put option: Right to sell an asset.
    - ➢If the exercise price is below the market price, the option is "out of the money" and will NOT be used.
- The value of an option depends on
  - ✓The value of the underlying variable
  - ✓The level of uncertainty
  - ✓The time to maturity
  - ✓Time value of money
  - ✓Counter-party risk

Types of Real Options



Early entrants can scale up later through cost-effective sequential investments as markets grow.

Speedy commitment to first generation of product or technology gives a firm preferential position to switch to next generation.

Investments in proprietary assets in one industry enable a firm to enter another industry cost effectively.

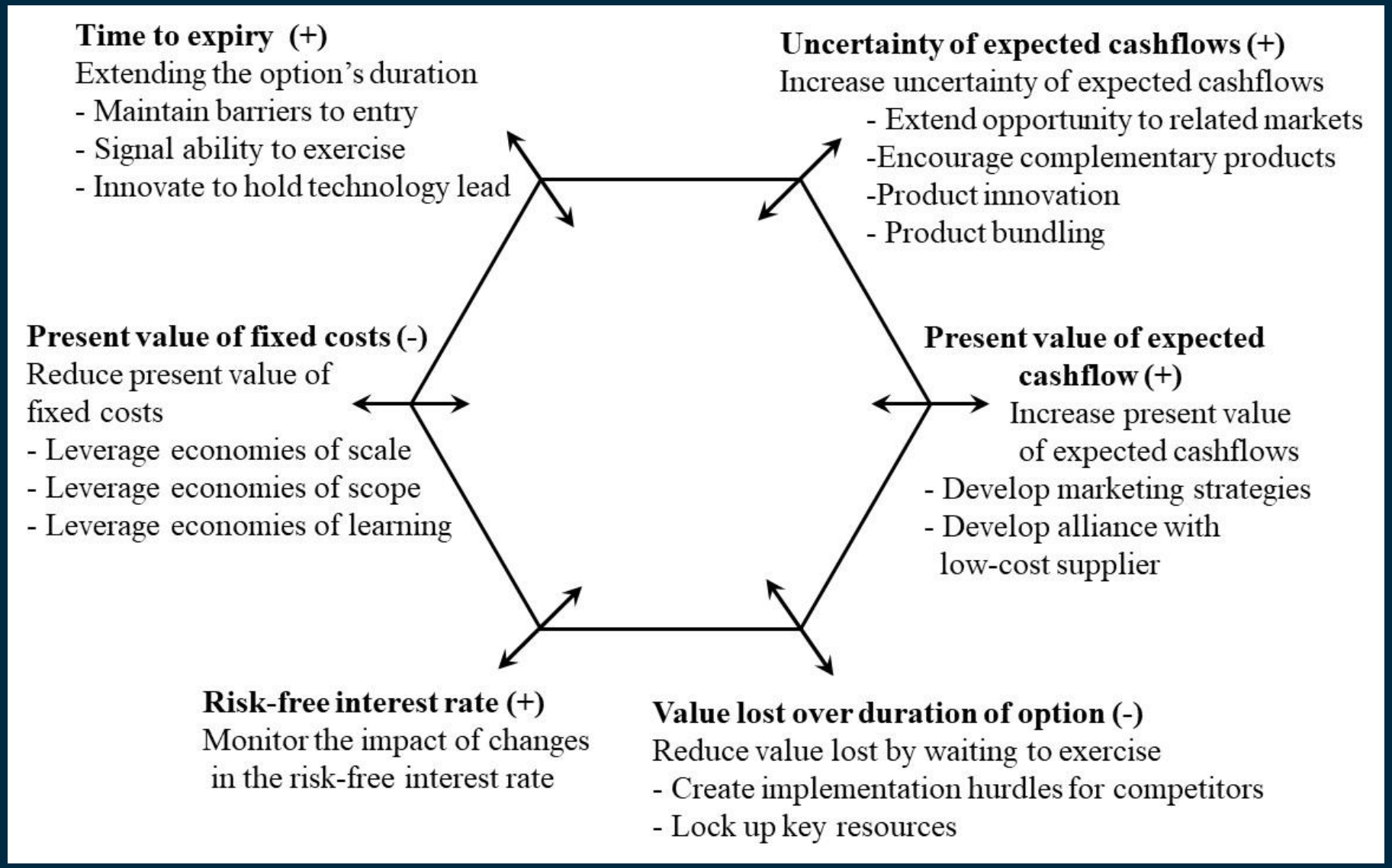
Delay investment until more information or skills are acquired.

Shrink or shut down a project part way through if new information changes the expected payoffs.

Limit the scope of (or abandon) operations when there are limited business opportunities.

Design projects so that they can be repurposed or switch to more cost-effective and flexible assets as new information is obtained.

Ways to increase the value of Real Options



Source: *Leslie, K.* and *Michaels, M.* (1997) The real power of real options, *McKinsey Quarterly*, 1997/3, 4–22, p.12.

Investment appraisal Open questions

- How do you establish an appropriate hurdle rate r?
  - ✓Cost of loan?
  - ✓Expected rate of return of shareholders?
  - ✓Cost of capital to the firm?

Durham University Business School

END

AACSB ACCREDITED · ASSOCIATION OF MBAS ACCREDITED · EFMD EQUIS ACCREDITED

LEADING BUSINESS THINKING

This Photo by Unknown Author is licensed under CC BY-NC-ND